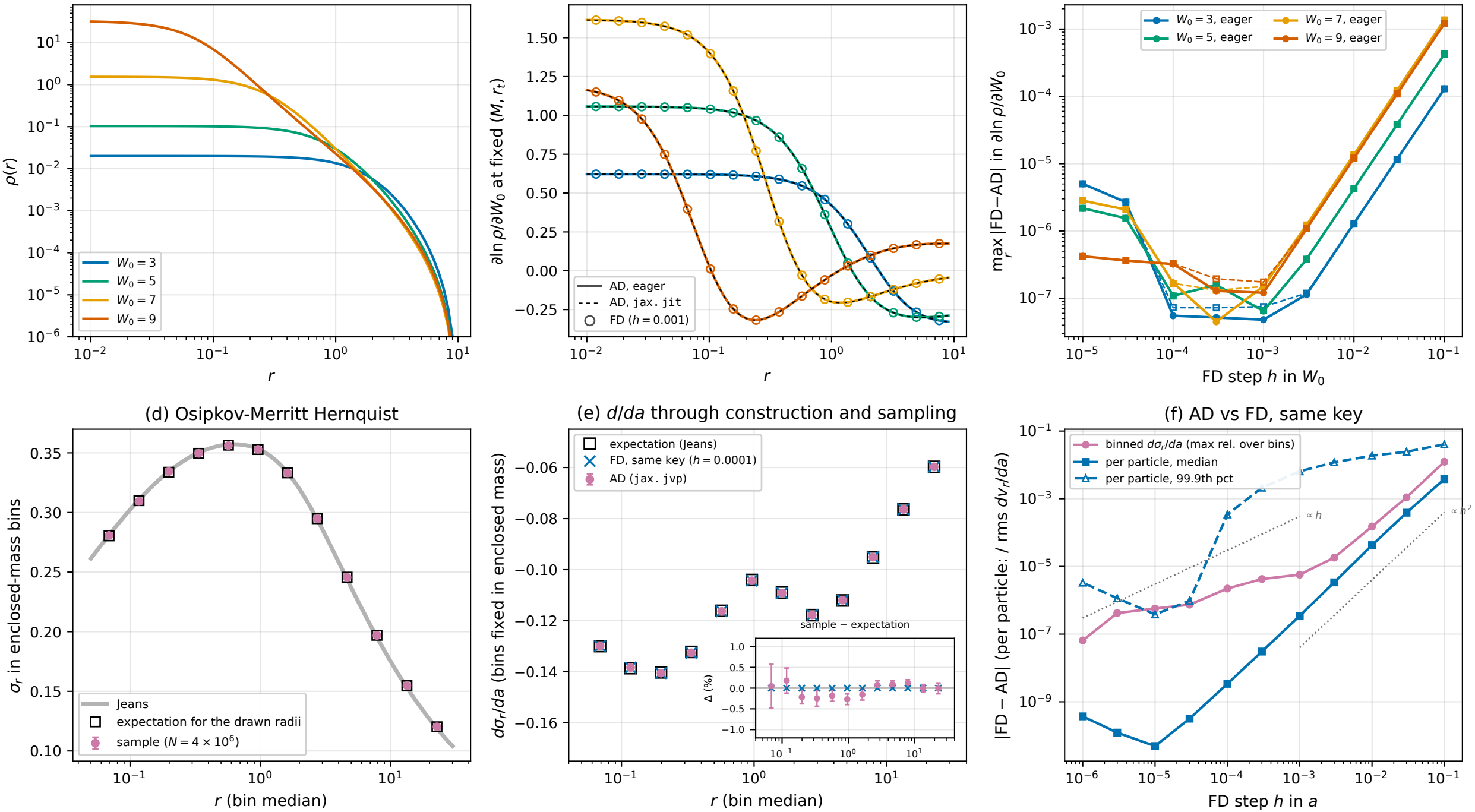

(a) King models
$\rho(r)$
r
$W_0 = 3$
$W_0 = 5$
$W_0 = 7$
$W_0 = 9$
(b) d/dW_0 through the ODE solve
$\partial \ln \rho / \partial W_0$ at fixed (M, r_t)
r
AD, eager
AD, jax.jit
FD ($h = 0.001$)
(c) AD vs FD: eager solid, jit dashed
$\max_r |\mathrm{FD} - \mathrm{AD}|$ in $\partial \ln \rho / \partial W_0$
FD step h in W_0
$W_0 = 3$, eager
$W_0 = 5$, eager
$W_0 = 7$, eager
$W_0 = 9$, eager
(d) Osipkov-Merritt Hernquist
σ_r in enclosed-mass bins
r (bin median)
Jeans
expectation for the drawn radii
sample ($N = 4 \times 10^6$)
(e) d/da through construction and sampling
$d\sigma_r/da$ (bins fixed in enclosed mass)
r (bin median)
expectation (Jeans)
FD, same key ($h = 0.0001$)
AD (jax.jvp)
sample − expectation
Δ (%)
(f) AD vs FD, same key
$|\mathrm{FD} - \mathrm{AD}|$ (per particle: / rms dv_r/da)
FD step h in a
binned $d\sigma_r/da$ (max rel. over bins)
per particle, median
per particle, 99.9th pct
$\propto h$
$\propto h^2$